Array

Fundamentals of Programming – CSC10012

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Contents

- Array 1-D.
- Array 2-D.

Array 1-D

- Consider a program:
  - Enter 5 integers, then print them out.

    Declare 5 ints a1, a2, a3, a4, a5.
  - Enter 50 integers, then print them out.

    Declare 50 ints!

  ➔ How to declare many variables at once?

  ➔ Use array.

Array 1-D

- Basic concepts:
  - A sequence of contiguous variables of the same type.
  - Each variable in array called array element.
  - Declaration:

<data type> **<array name>[** <array length> **]**;

<array length>: number of elements, must be constant.

```
int     m1[ 10 ];     // Array of 10 integers.
float   m2[ 50 ];     // Array of 50 floats.

int     N;
float   m3[ N ];      // Wrong.

const int K = 100;
float   m4[ K ];      // Array of 100 floats.
```

Array 1-D

- Basic concepts:
  - Uninitialized array has random element values.

int **m[5]**; **m**

?	?	?	?	?

  - Initialization:

<data type> **<array name>[**<array length>**]** = { **<value 1>**, **<value 2>**, ... };

int **m1[5]** = { 1, 2, 3, 4, 5 }; **m1**

1	2	3	4	5

// Initialize all elements.

int **m2[5]** = { 1, 2 }; **m2**

1	2	0	0	0

// Initialize some elements.
// remain elements = 0

int **m3[5]** = { 0 }; **m3**

0	0	0	0	0

// Initialize all elements = 0

int **m4[ ]** = { 1, 2, 3, 4, 5 }; **m4**

1	2	3	4	5

// Initialize with auto length

Array 1-D

- Basic concepts:
  - Accessing element:

    <array name> **[<index>]**

    <index>: integer in range **[0.. <array length> - 1]**.

```
int a[ 10 ] = { 0 };
```

	0	1	2	3	4	5	6	7	8	9
a	0	0	0	0	0	0	0	0	0	0

```
a[ 0 ] = 5;
a[ 1 ] = 6;
a[ 2 ] = a[ 0 ] + a[ 1 ];

a[ -1 ] = 7;        // Wrong.
a[ 10 ] = 8;        // Wrong.
```

Array 1-D

- Basic concepts:
  - Passed as argument:

    Method 1: void foo(**int a[100], int size**);

    ➔ Static array, accept only array of 100 elements.

    Method 2: void foo(**int a[ ], int size**);

    ➔ Dynamic array, accept array of any length.

    Array elements can be CHANGED.

```
void foo( int a[ ], int size )
{
    a[ 2 ] = 9;
    a[ 5 ] = 8;
}
```

```
int main()
{
    int a[ 100 ] = { 0 };
    int b[ 200 ] = { 0 };
    foo( a, 50 );
    foo( b, 70 );
    // a[2], a[5] changed.
    // b[2], b[5] changed.
}
```

Array 1-D

- Operations:
  - Step 1: Iterate through elements.

    Use **loop** + **counter**.
  - Step 2: Access element by indexing.

```
// Iterate array A length N.
for ( int i = 0; i < N; i++ )
{
    <Process element A[ i ]>;
}
```

Array 1-D

- Operations:

```
void inputArray( int a[ ], int &n )
{
        printf("Number of elements = ");
        scanf("%d", &n);

        for (int i = 0; i < n; i++)
        {
                printf("Element %d = ", i);
                scanf("%d", &a[ i ]);
        }
}
void printArray( int a[ ], int n )
{
        for ( int i = 0; i < n; i++ )
                printf("%d ", a[ i ]);
}
```

```
#define MAX     1000

int main()
{
        int  m[ MAX ], size;

        inputArray(m, size);
        printArray(m, size);
}
```

Array 1-D

- Operations:

```
// Sum of elements in a length n.
long sumArray( int a[ ], int n )
{
        long  sum = 0;

        for ( int i = 0; i < n; i++ )
                sum += a[ i ];

        return sum;
}
```

```
#define MAX      100

int main()
{
        int  m[ MAX ], size;

        inputArray(m, size);
        long  tong = sumArray(m, size);
}
```

Array 1-D

- Operations:
  - Copy array:

```
int  a [ MAX ] = { 1, 2, 3, 4, 5 };
int  b [ MAX ];
b = a;                          // Wrong, cannot copy by assignment!
copyArray( a, n, b );           // Write function to copy a length n to b.
```

  - Insert element:

```
int  a [ MAX ] = { 1, 2, 4, 5 };
int  x = 3;
int  pos = 2;                    // Array a length n, insert x at index pos.
insertArray( a, n, x, pos );  // a = { 1, 2, 3, 4, 5 }
```

  - Delete element:

```
int  a [ MAX ] = { 1, 2, 3, 4, 5 };
int  pos = 2;                   // Array a length n, delete element at index pos.
deleteArray( a, n, pos );    // a = { 1, 2, 4, 5 }
```

Array 1-D

- Array of struct:

```
struct Student  m1[ 10 ];                    // Array of 10 struct Student.

struct Student  m2[ 10 ] = {                 // Initialize some array elements.
    { “24127001“, “minh”, 8.5, 9.0 },
    { .name = “trang”, .literature = 9.5 }
};

struct Student  m2[  ] = {                   // Auto array length.
    { “24127001”, “minh”, 8.5, 9.0 },
    { .name = “trang”, .literature = 9.5 }
};
```

Contents

- Array 1-D.
- Array 2-D.

Array 2-D

- Consider the program:
  - Enter a matrix of 5 x 10 integers, then print it out.

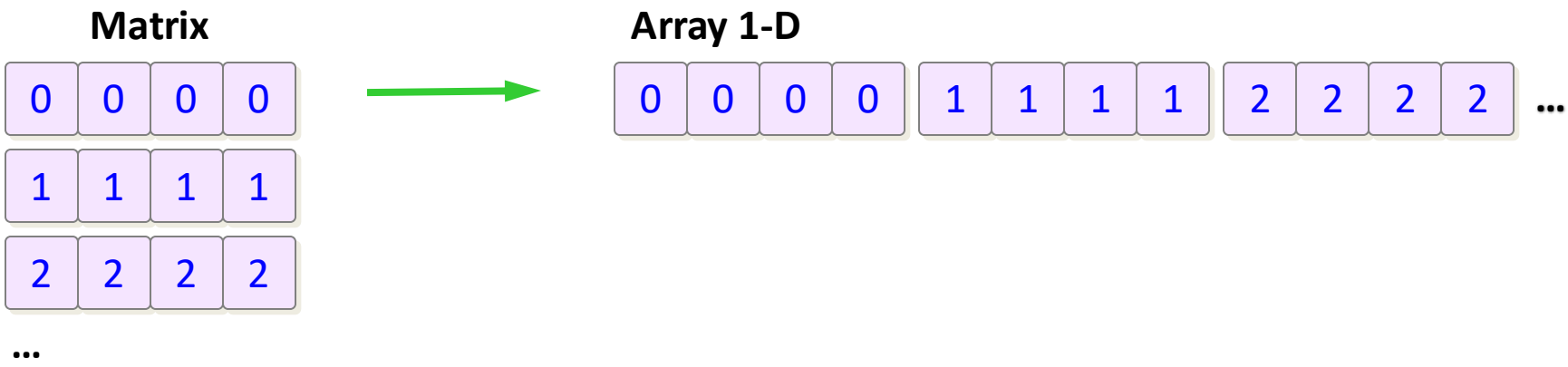
    Declare 5 arrays: int a1[10], a2[10], a3[10], a4[10], a5[10].
  - Enter a matrix of 50 x 10 integers, then print it out.

    Declare 50 arrays!

  ➔ How to declare a matrix of M x N?

Array 2-D

- Method 1:
  - Use array 1-D!
  - Matrix M x N ~ array 1-D of M x N elements.
  - Matrix [row i, column j] ~ array [i * N + j].



Array 2-D

- Method 2:
  - Use array 2-D.
  - Declaration:

    <data type> **<array name>[**<rows>**] [**<columns>**]**;

    <rows>, <columns> must be constants.

    int **m1[5][10]**; // Matrix 5 x 10 of integers.

    int **m2[M][N]**; // Wrong.
  - Accessing element:

    <array name> **[<row index>] [<column index>]**

    <row index>: integer in range **[0.. <rows> - 1]**.

    <column index>: integer in range **[0.. <columns> - 1]**.

    m1**[0][2]** = 5;

    m1**[1][3]** = 6;

    m1**[-1][10]** = 7; // Wrong.

Array 2-D

- Initialization:

```
<data type> <array name>[<rows>] [<columns>] =
{
  <Initialize row 0>,
  <Initialize row 1>,
  ...
};
```

```
// Initialize all elements.
int m1[ 3 ][ 5 ] =
{
        { 1, 1, 1, 1, 1 },
        { 1, 2, 3, 4, 5 },
        { 5, 4, 3, 2, 1 }
};
```

```
// Initialize some elements.
int m1[ 3][ 5 ] =
{
        { 1, 1 },
        { 1, 2, 3 },
        { 0 }
};
```

```
// Initialize with auto rows.
int m1[  ][ 5 ] =
{
        { 1, 1 },
        { 1, 2, 3 },
        { 0 }
};
```

Array 2-D

- Operations:
  - Step 1: Iterate through array.

    Use **2 nested loops**.

    Outer loop: iterate through rows.

    Inner loop: iterate through columns in each row.
  - Step 2: Access element by indexing.

```
// Iterate matrix A of M rows N columns.
for ( int i = 0; i < M; i++ )
    for ( int j = 0; j < N; j++ )
    {
        <Process element A[ i ][ j ]>;
    }
```

Array 2-D

- Operations:

```
// Input matrix A of M rows N columns.
void inputMatrix( int A[ ][ MAX ], int &M, int &N )
{
    printf("Enter rows, columns = ");
    scanf("%d %d", &M, &N);

    for (int i = 0; i < M; i++)
      for (int j = 0; j < N; j++)
      {
          printf("Element %d, %d = ", i, j);
          scanf("%d", &A[ i ][ j ]);
      }
}
```

```
#define MAX     100

int main()
{
        int  a[ MAX ][ MAX ];
        int  M, N;

        inputMatrix(a, M, N);
}
```

Summary

- Array 1-D:
  - Sequence of contiguous elements of same type.
  - Element accessed by indexing.
  - Operations: use loop to iterate through elements.
- Array 2-D:
  - Method 1: array 1-D of rows x columns elements.
  - Method 2: array 2-D.
  - Matrix of elements of same type.
  - Operations: use nested loop to iterate.

Practice

- Practice 5.1:

Write C/C++ program (organize in functions and multiple-file project):

- Enter an array of integers.
- Count:
  - a) Negative numbers in the array.
  - b) Prime numbers in the array.

Input format:

Enter N = 3

Element 0 = 2

Element 1 = 3

Element 2 = -6

Output format:

Negative numbers: 1.

Prime numbers: 2.

Practice

- Practice 5.2:

Write C/C++ program to check array:

(organize in functions and multiple-file project)

- Enter a array of integers.
- Check if:

a) Array is in ascending order.

b) Array is symmetric.

c) Array is an arithmetic progression.

Output format:

Array <is/is not> ascending.

Array <is/is not> symmetric.

Array <is/is not> a arithmetic progression.

Practice

- Practice 5.3:

  Write C/C++ program to operate on students:

  (organize in functions and multiple-file project):

  - Declare struct to represent a student (stated in the lesson).
  - Enter a list of N students.
  - Print a list of excellent students (GPA >= 8.5) in descending order of GPA.

Practice

- Practice 5.4:

  Write C/C++ program to operate on date:

  (organize in functions and multiple-file project):

  - Declare struct to represent date (day, month, year).
  - Enter a list of date.
  - Print the list from the latest date to the oldest one.

Practice

- Practice 5.5:

Write C/C++ program as follow:

(organize in functions and multiple-file project)

- Enter square matrix N x N of integers.
- Print to screen:
  - a) Sum of elements on each diagonal.
  - b) Row index having max sum of elements.
  - c) It is a magic square or not.

Output format:

Main diagonal = <sum of elements>.

Anti-diagonal = <sum of elements>.

It <is/is not> a magic square.

Practice

- Practice 5.6:

  Write C/C++ program to manipulate matrix:

  (organize in functions and multiple-file project)

  - Enter a matrix M x N of integers.
  - Print to screen all elements satisfying its value equals to sum of the remaining elements on its row and column.

Practice

- Practice 5.7:

  Write C/C++ program to rotate matrix:

  (organize in functions and multiple-file project)

  - Enter a matrix M x N of integers.
  - Rotate left matrix 90 degree and print result.
  - Rotate right matrix 90 degree and print result.

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&
A